

Quantum Intelligence System (QIS)

PointClickCare Developer Program — Workflow Review Submission

APPLICATION ID
quantumintelligence

Organization	Apex Healthcare, LLC	Organization ID	apequa2310
Submission Date	2026-05-23	PCC Base URL	https://connect.pointclickcare.com
Backend Stack	Python 3.11 / FastAPI / Railway (cloud)	Facilities	Multi-Medical Center (ID 324, ~110 res.) · Heritage Center (ID 327, ~142 res.)

§1 APPLICATION OVERVIEW

The **Quantum Intelligence System (QIS)** is a secure, HIPAA-compliant clinical intelligence platform developed exclusively for Apex Healthcare, LLC. QIS integrates with PointClickCare via OAuth2 to retrieve real-time census and patient data across two skilled nursing facilities. It applies a proprietary 10-domain clinical risk engine to generate composite risk scores for each resident, surfaces AI-assisted care plan suggestions for clinician review, and presents a unified triage dashboard to support daily clinical decision-making.

All AI-generated care plan language is presented to the clinician for review and explicit approval before any write-back to PCC. No PHI is persisted beyond the active session. The system operates under a signed BAA with Anthropic (Claude API) and follows PointClickCare API usage policies.

§2 INTENDED USERS & ROLES

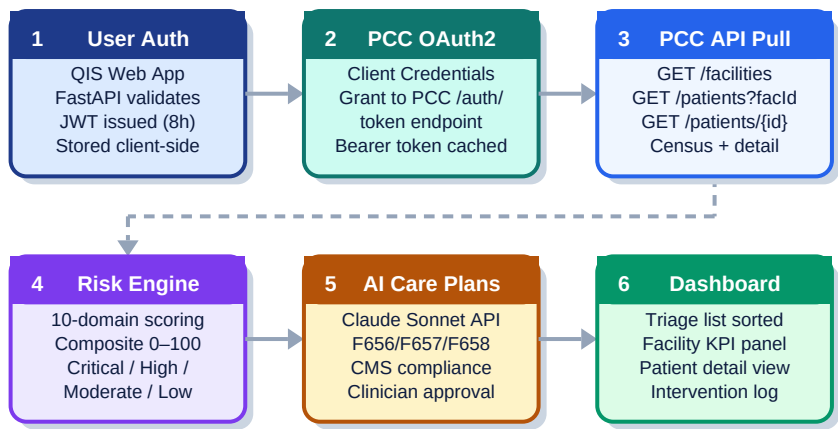
Medical Director	Admin	Clinical triage review · Risk score investigation · Intervention logging · Care plan review & approval
Director of Nursing (DON)	Admin	Census monitoring · High-risk resident alerts · Daily triage prioritization · Staff assignment support
Facility Administrator	Viewer	Operational KPI dashboard · Facility-level risk distribution · Census counts · Non-clinical summary views

§3 PCC API ENDPOINTS CALLED

POST	/auth/token	Client Credentials Grant — obtain Bearer token for PCC API access	Customer Key + Client Secret
GET	/api/public/preview1/facilities	Retrieve list of authorized facilities for the organization	Bearer token

GET	/api/public/preview1/patients?facId={id}	Retrieve patient census for a given facility (IDs 324, 327)	Bearer token
GET	/api/public/preview1/patients/{patientId}	Retrieve individual patient record for risk engine input	Bearer token

§4 BACK-END WORKFLOW DIAGRAM



Data flow: User Auth → PCC OAuth2 Token → PCC API Data Pull → Clinical Risk Engine → AI Care Plans Layer → Dashboard Rendering

§5 DETAILED BACK-END WORKFLOW

Step 1 — User Authentication

- User accesses the QIS web application (hosted on Railway).
- User submits credentials (username + password) via HTTPS POST.
- FastAPI validates credentials against the user database (bcrypt hash comparison).
- On success, a JWT is issued with 8-hour expiry and HS256 signing.
- JWT is stored client-side (memory / httpOnly cookie) and included as Authorization: Bearer on all subsequent API requests.

Step 2 — PCC OAuth2 Token Acquisition

- QIS backend performs a Client Credentials Grant (OAuth2) to <https://connect.pointclickcare.com/auth/token>.
- Request includes: grant_type=client_credentials, client_id (Customer Key: pg9JCok3xQQo9bp0cAEGbbiAtfJkWsxY), and client_secret.
- PCC returns a Bearer access token with a defined expiry window.
- Token is cached in memory with expiry tracking; auto-refreshed before expiry without user interaction.

Step 3 — PCC API Data Pull

- GET /api/public/preview1/facilities — retrieves authorized facility roster.
- GET /api/public/preview1/patients?facId=324 — pulls census for Multi-Medical Center (~110 residents).
- GET /api/public/preview1/patients?facId=327 — pulls census for Heritage Center (~142 residents).
- GET /api/public/preview1/patients/{patientId} — fetches individual patient detail records as needed for risk scoring.
- All calls include Authorization: Bearer header. Responses are parsed and held in session memory only.

Step 4 — Clinical Risk Engine (10-Domain Scoring)

- Each patient record is evaluated across 10 clinical domains:
- (1) Fall Risk · (2) Wound/Skin Integrity · (3) Infection/Sepsis Risk
- (4) Nutrition/Weight Status · (5) Pain Management · (6) Cognition/Behavioral
- (7) Functional Decline · (8) Readmission Risk · (9) Medication Safety · (10) Goals of Care
- Domain scores are weighted and summed into a Composite Score (0–100).
- Risk bands: Critical (80–100) · High (60–79) · Moderate (40–59) · Low (0–39).
- Triage list is sorted descending by composite score for clinical review.

Step 5 — AI Care Plans Layer (Claude Sonnet API)

- Care plan text (from PCC or clinician input) is submitted to the Claude Sonnet API.
- Prompt instructs the model to evaluate compliance with CMS regulations: F656 (care plan development), F657 (care plan revision), F658 (services provided meet professional standards).
- Model returns: compliance assessment, identified gaps, and SMART-goal rewrites.
- Output is displayed to the clinician in a review panel — no automatic write-back occurs.
- Clinician explicitly approves, edits, or discards AI suggestions before any changes are submitted to PCC.
- BAA with Anthropic is in place; no PHI is retained by the Claude API beyond the call.

Step 6 — Dashboard Rendering & Output

- Triage list rendered sorted by composite risk score (Critical first).
- Facility KPI panel: risk distribution chart, census counts, average composite score per facility.
- Patient detail view: demographics, risk domain breakdown, care plan status, clinical narrative.
- Printable patient reports generated as PDF exports for rounds/meetings.
- Intervention logging: clinicians log actions taken; entries persisted to PostgreSQL database with timestamp and clinician ID.
- All dashboard data is derived from in-session PCC API responses; no PHI is written to persistent storage beyond the intervention log.

§6 DATA SECURITY & COMPLIANCE

Transport Security	All connections use TLS 1.2+ (HTTPS). Railway enforces HTTPS on all endpoints.
User Authentication	JWT tokens (HS256, 8-hour expiry). Passwords stored as bcrypt hashes. No plaintext credentials stored.
PCC Token Handling	PCC Bearer token cached in server memory only. Never written to disk, logs, or database. Auto-refreshed on expiry.
PHI Storage Policy	No PHI persisted beyond the active session. PCC patient data lives only in server memory during API response processing.
Intervention Log	Only clinician-authored free-text notes, timestamps, and clinician IDs are written to the database. No PCC patient fields stored.
AI/LLM (Claude API)	BAA signed with Anthropic. Care plan text is submitted per-call; Anthropic does not retain PHI beyond the API response.

Access Control	Role-based: Admin (Medical Director, DON) vs. Viewer (Administrator). Enforced at API layer via JWT claims.
Audit & Logging	Application-level access logs retained on Railway. No PHI included in logs. Intervention actions logged with actor identity.
BAA Coverage	Signed BAAs in place: Anthropic (Claude) ✓ · Google ✓ · OpenAI ✓ · Railway (infra) ✓